# Mediana a doi vectori sortaţi

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Se dau doi vectori a și b de lungime n, cu elementele ordonate crescător. Să se determine mediana vectorului obținut prin interclasarea celor doi vectori.

# Mediana a doi vectori sortați

Exemplu: n = 5

1 12 15 16 38

2 13 17 30 45

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Exemplu: n = 5

1 12 15 16 38

2 13 17 30 45



1 2 12 13 15 16 17 30 38 45

Mediana (15+16)/2 = 15,5

# Mediana a doi vectori sortați

- **Algoritm O(n)** – interclasăm vectorii şi apoi aflăm mediana în timp constant (din elementele de la mijlocul vectorului, conform definiţiei)

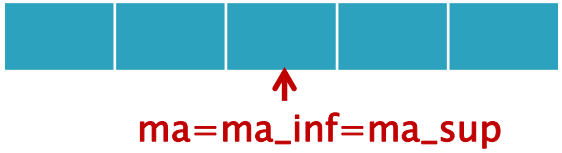
# Mediana a doi vectori sortați

- **Algoritm O(log(n))**

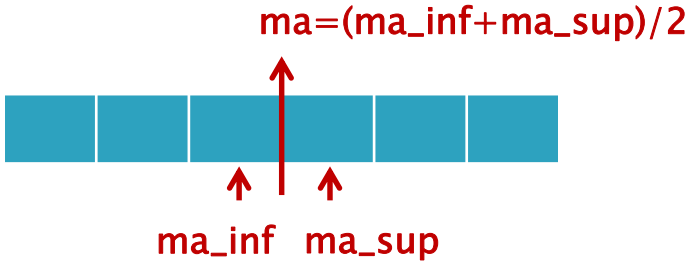
# Mediana a doi vectori sortați

- Fie **`ma_inf, ma_sup, ma`** mediana inferioară, superioară, respectiv mediana vectorului `a`
- **`mb_inf, mb_sup, mb`** – similar pentru vectorul `b`
- **`c`** – vectorul obținut prin interclasare

- n impar:




- n par:



# Mediana a doi vectori sortați



- Comparăm ma şi mb

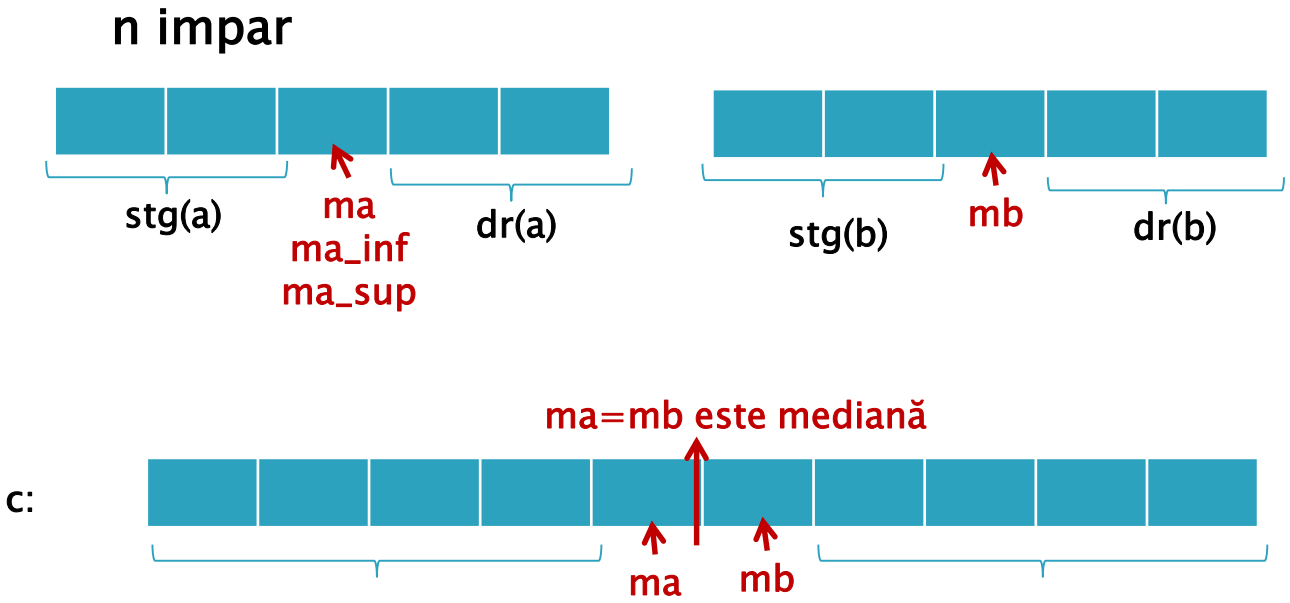


# Mediana a doi vectori sortați

- $ma = mb \Rightarrow mc = ma = mb$?

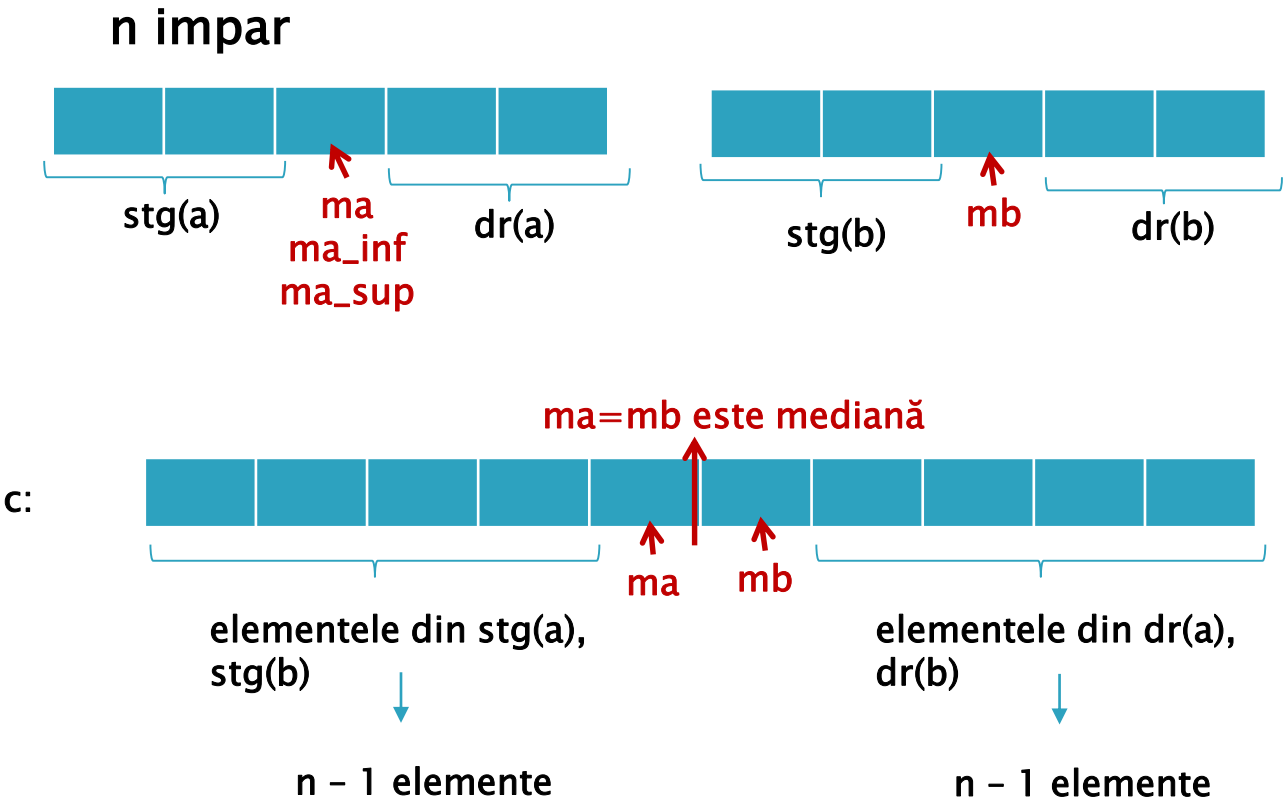
# Mediana a doi vectori sortați

- ma = mb ⇒ mc = ma = mb



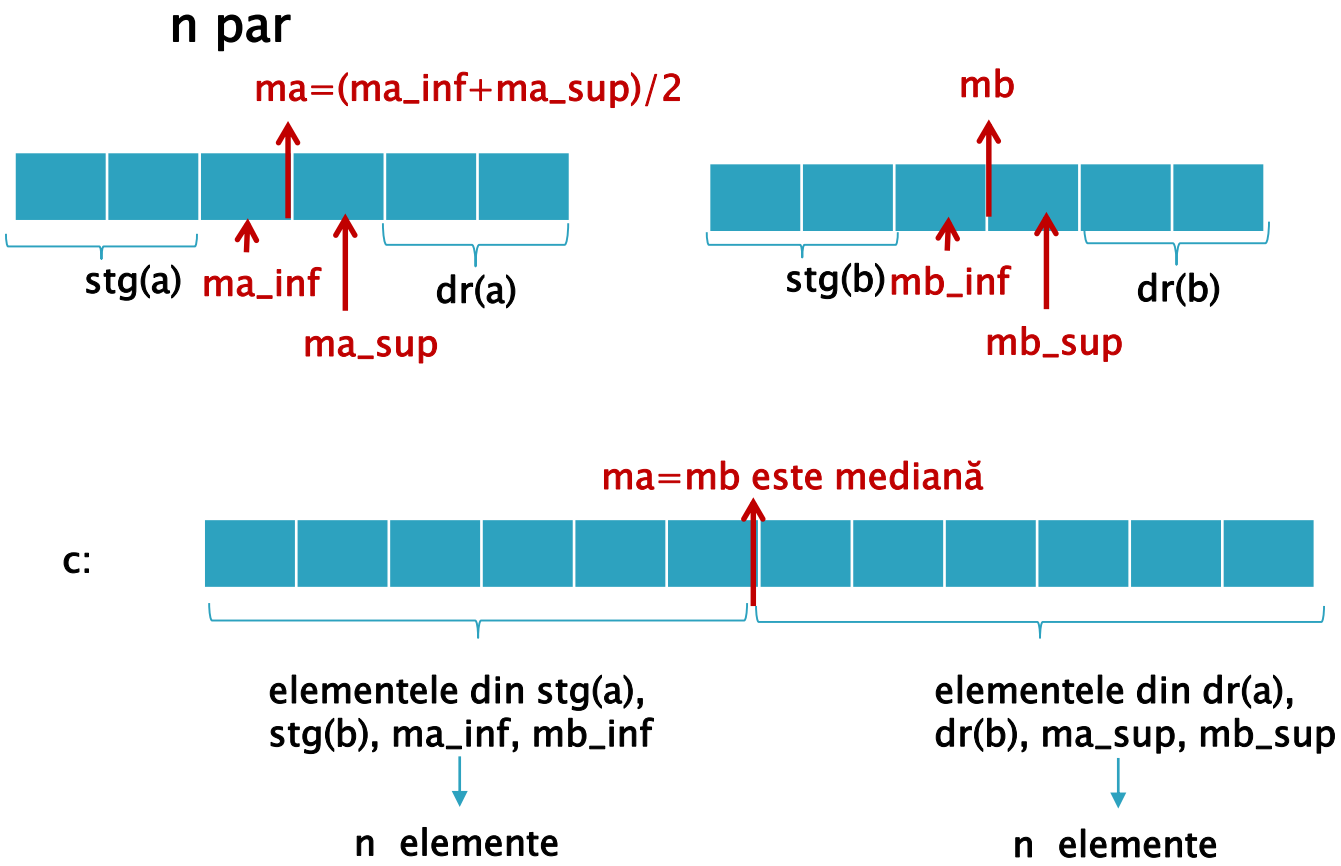
# Mediana a doi vectori sortați

- $ma = mb \Rightarrow mc = ma = mb$



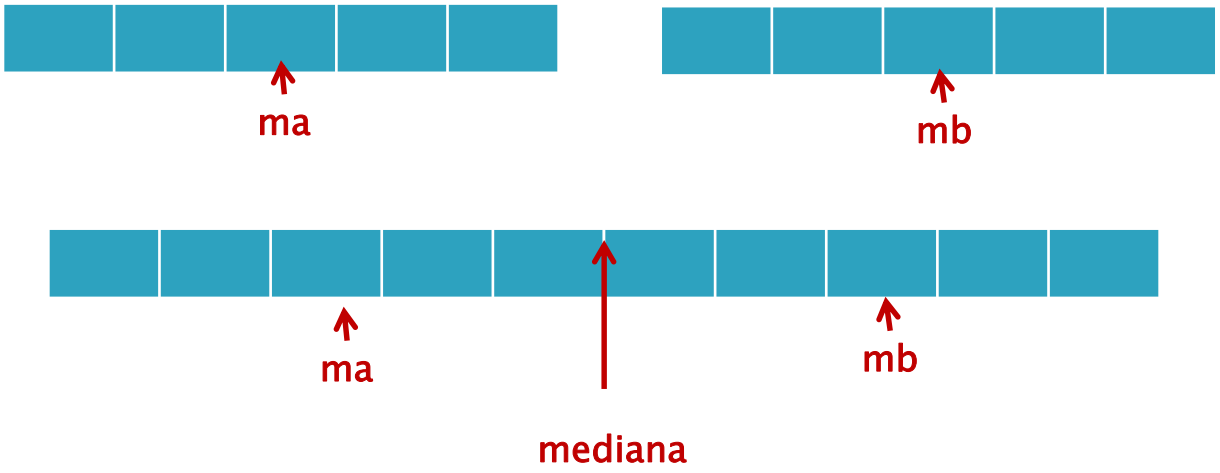
# Mediana a doi vectori sortați

- $ma = mb \Rightarrow mc = ma = mb$



# Mediana a doi vectori sortați

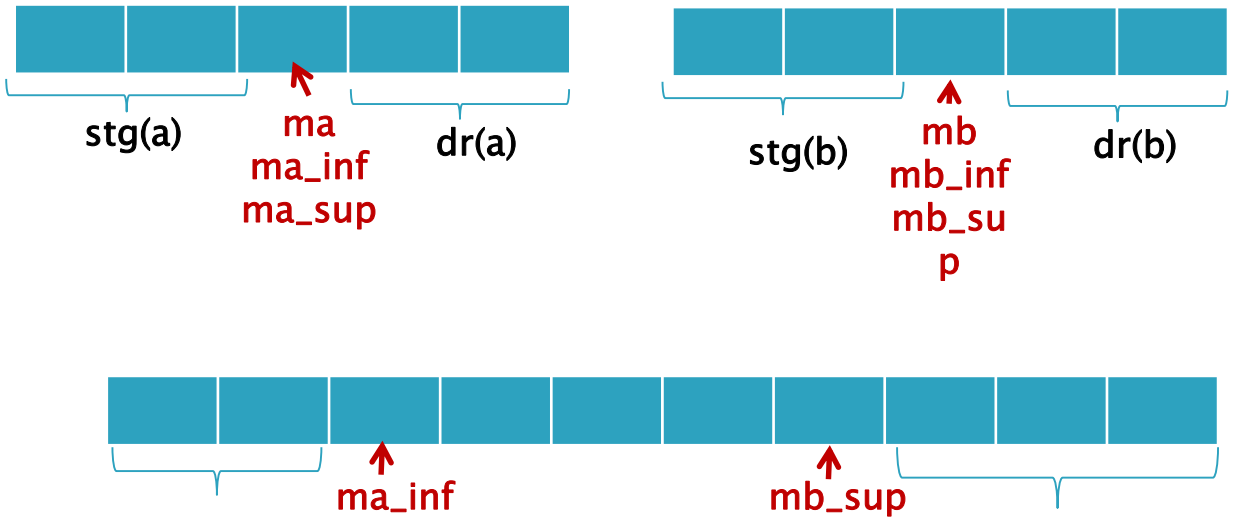
- $ma < mb \Rightarrow mc \in [ma, mb]$?



# Mediana a doi vectori sortați

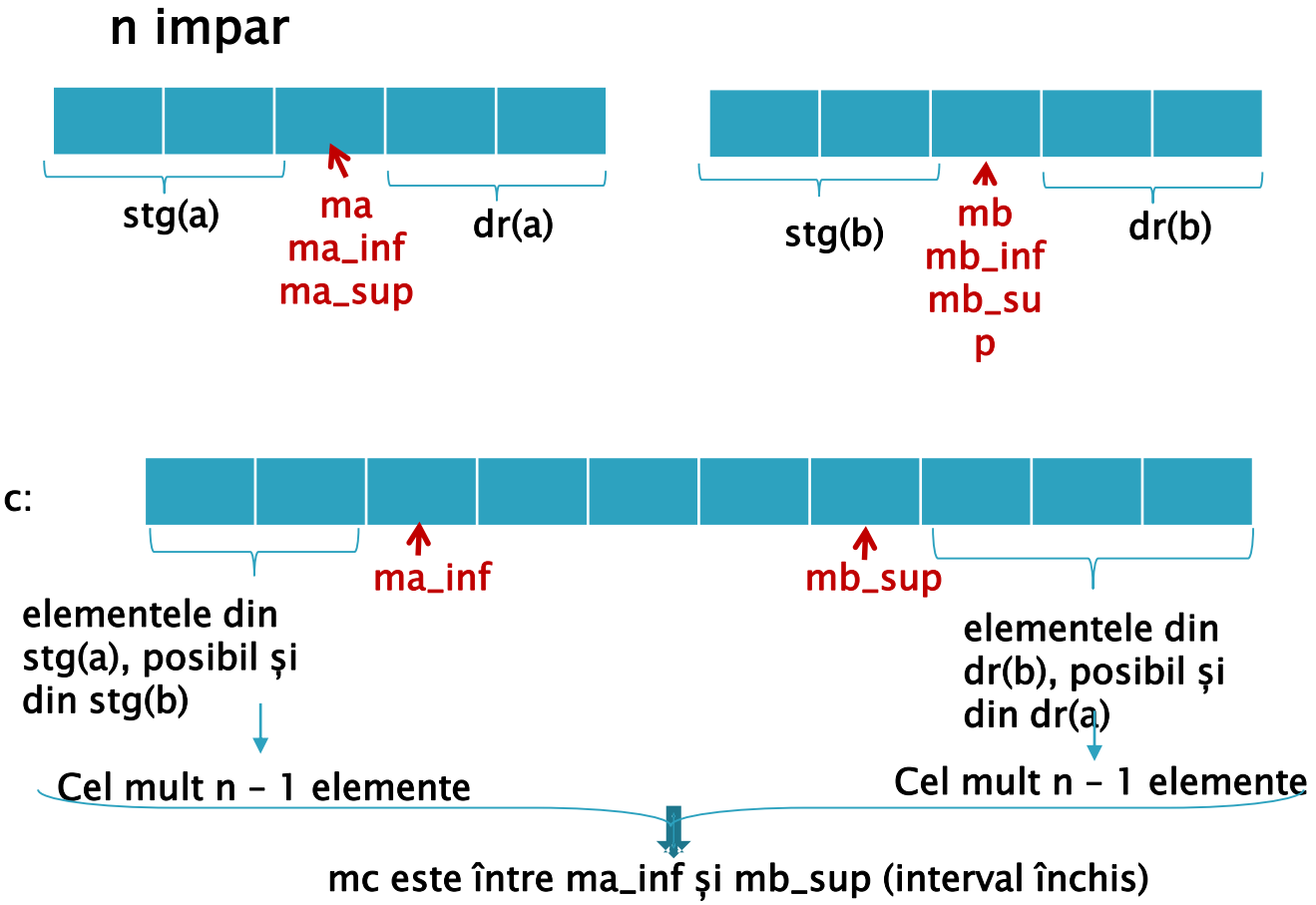
- $ma < mb \Rightarrow mc, mc_inf, mc_sup \in [ma_inf, mb_sup]$

**n impar**



# Mediana a doi vectori sortați

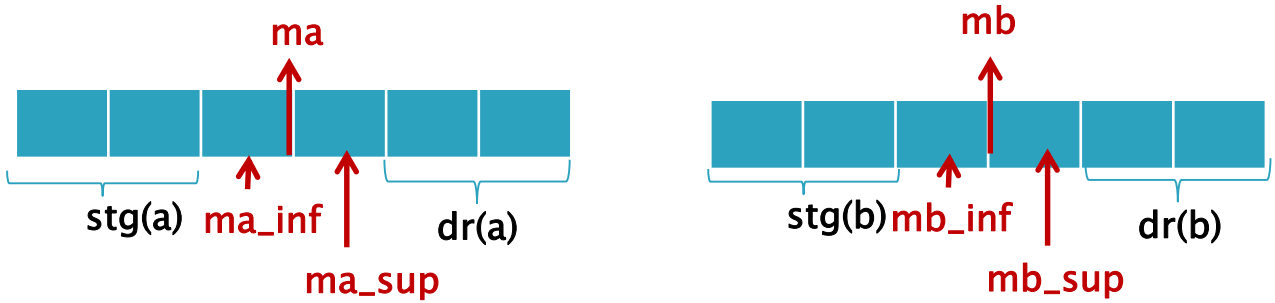
- $ma < mb \Rightarrow mc, mc_inf, mc_sup \in [ma_inf, mb_sup]$



# Mediana a doi vectori sortați

- $ma < mb \Rightarrow mc, mc_inf, mc_sup \in [ma_inf, mb_sup]$

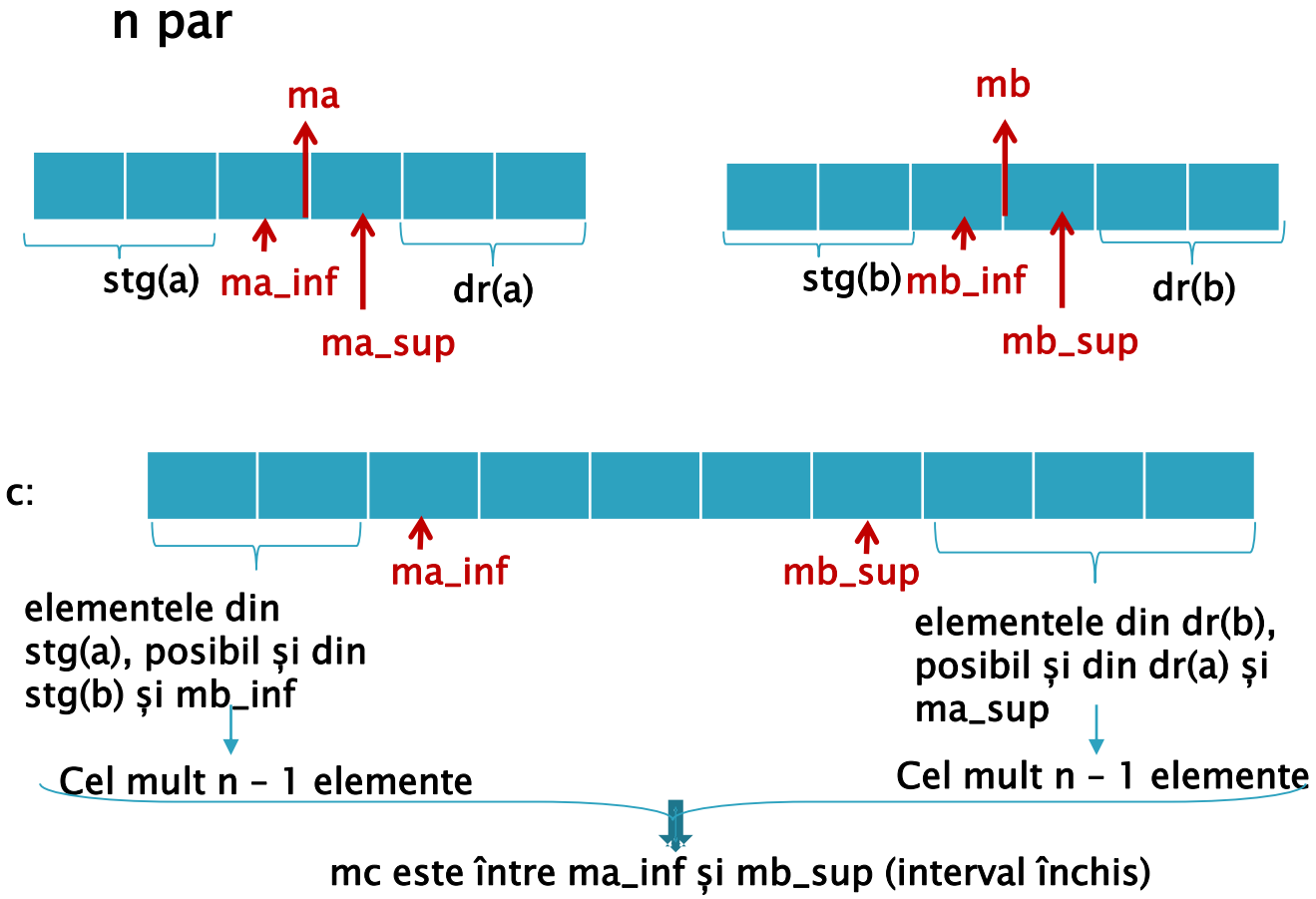
**n par**






# Mediana a doi vectori sortați

- $ma < mb \Rightarrow mc, mc_inf, mc_sup \in [ma_inf, mb_sup]$



# Mediana a doi vectori sortați

## Rezultă:

Pentru a determina mediana este suficient să considerăm:

- Subvectorul drept din primul vector ( inclusiv mediana inferioară)
- Subvectorul stâng din al doilea vector (inclusiv mediana superioară)

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## Rezultă:

Pentru a determina mediana este suficient să considerăm:

- Subvectorul drept din primul vector ( inclusiv mediana inferioară)
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Astfel

- din vectorul a renunțăm la [(n-1)/2] elemente care sunt înaintea lui ma_inf (deci și a lui mc_inf) în c
- din vectorul b renunțăm **tot la** [(n-1)/2] elemente care sunt după lui mb_sup (deci și după mc_sup) în c

deci **media noilor vectori este egală cu mediana lui c**

- `ma > mb` - Similar

# Mediana a doi vectori sortați

Corectitudine:

**mediana noii probleme = mediana problemei inițiale**

# Pseudocod

# Mediana a doi vectori sortați

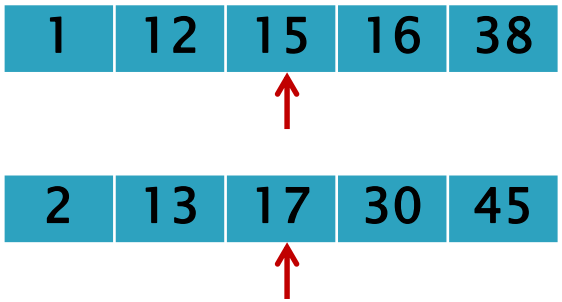
- Fie `ma` mediana vectorului `a` și `mb` mediana vectorului `b`
  - Dacă `ma = mb` atunci **această valoare este mediana**

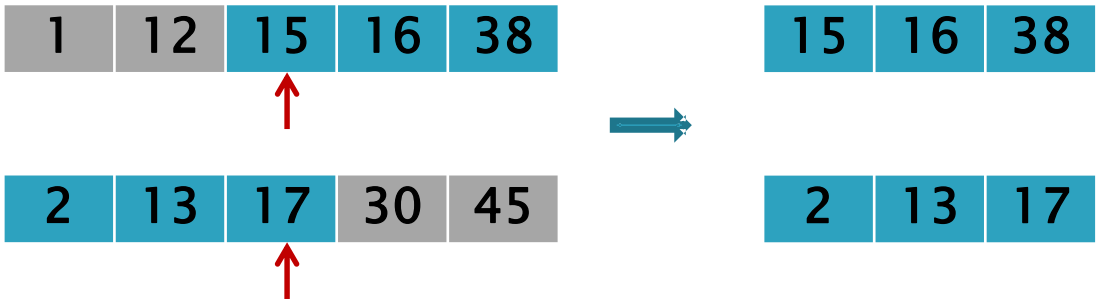
# Mediana a doi vectori sortați

- Fie `ma` mediana vectorului `a` și `mb` mediana vectorului `b`
  - Dacă `ma = mb` atunci **această valoare este mediana**
  - Dacă `ma > mb` atunci mediana = mediana **subvectorilor**

    `a[ 0..⌊n/2⌋ ], b[ ⌊(n-1)/2⌋..n-1 ]`
  - Dacă `ma < mb` atunci mediana = mediana **subvectorilor**

    `a[ ⌊(n-1)/2⌋..n-1 ], b[ 0..⌊n/2⌋ ]`


1 12 15 16 38
2 13 17 30 45


1 12 15 16 38
2 13 17 30 45
15 16 38
2 13 17

# Mediana a doi vectori sortați

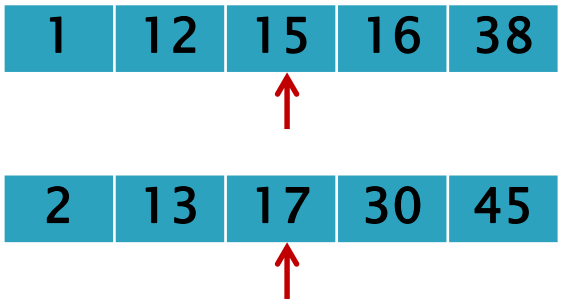
- **Știm să rezolvăm direct**:
  - n = 1: (a[1]+b[1])/2
  - n = 2: mediana lui {a[1],b[1], a[2],b[2]}

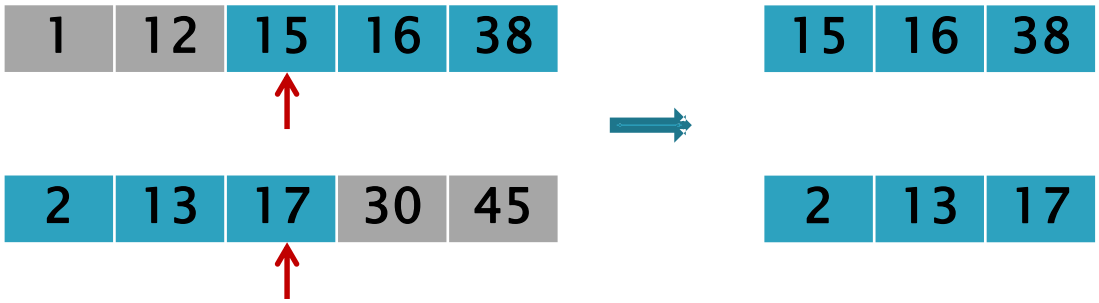
    = (max{a[1],b[1]}+min{a[2],b[2]})/2

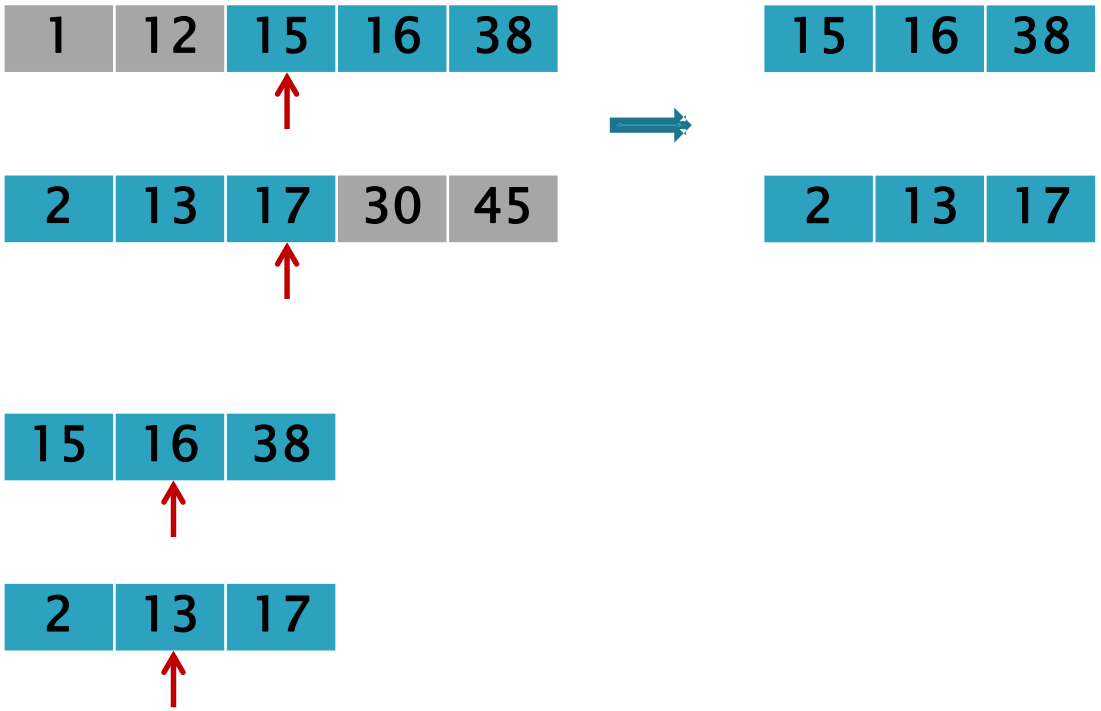
# Exemplu

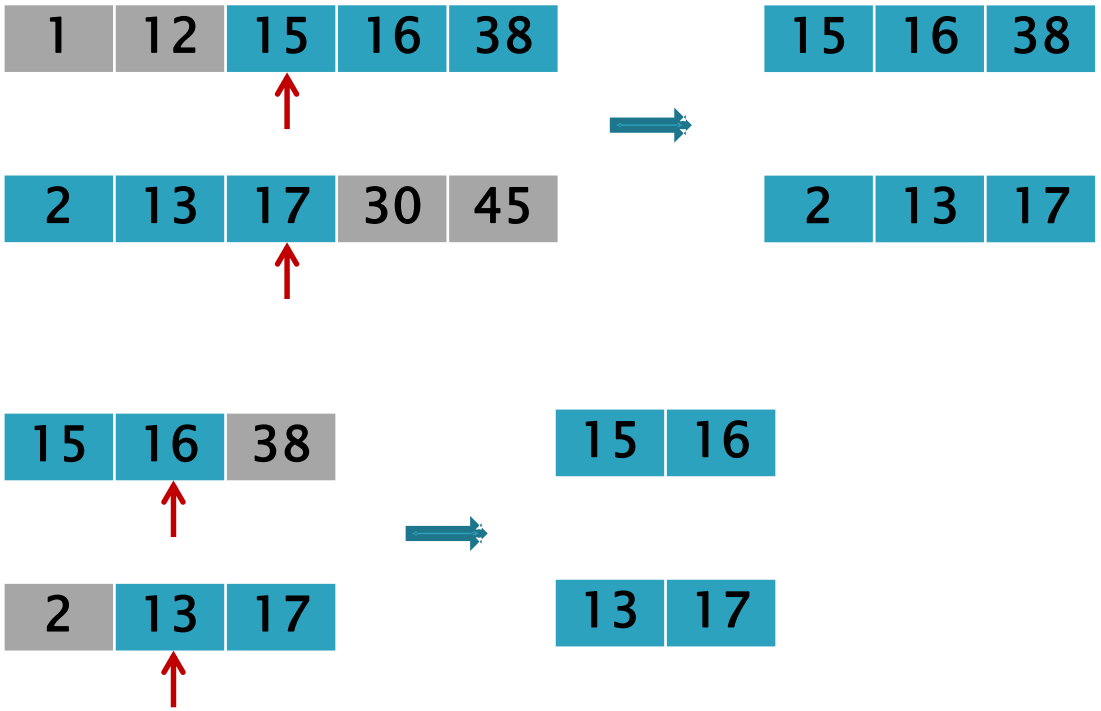
| 1 | 12 | 15 | 16 | 38 |
|---|---|---|---|---|

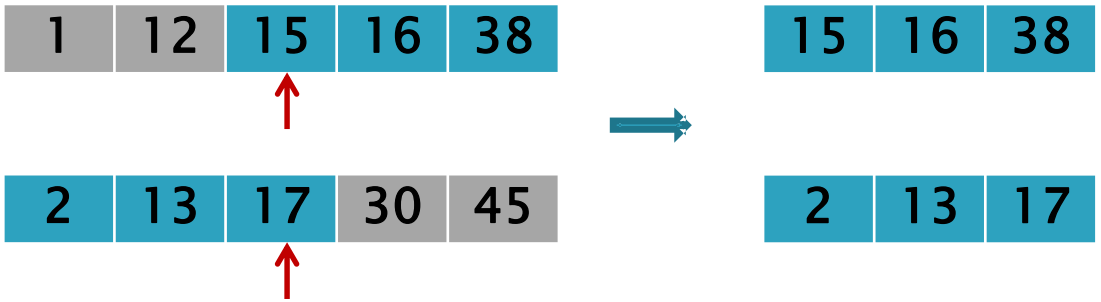
| 2 | 13 | 17 | 30 | 45 |
|---|---|---|---|---|

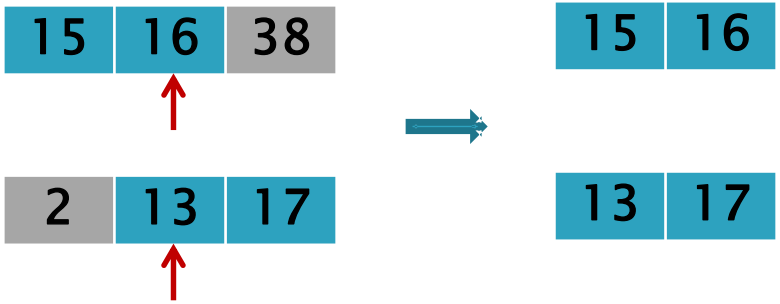

1 12 15 16 38
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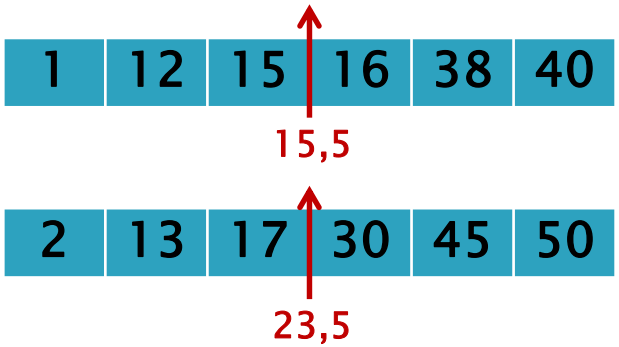

$$\text{Mediana} = \frac{\max\{13,15\}+\min\{16,17\}}{2} = \frac{15+16}{2}$$

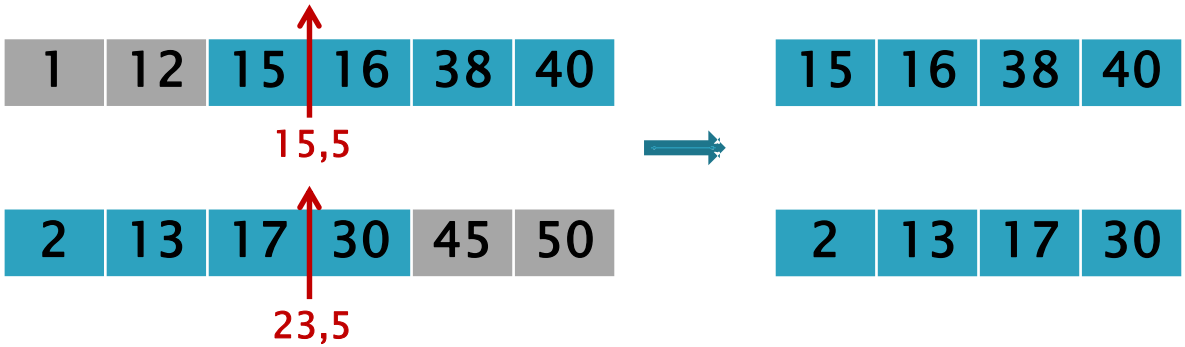
$$= 15{,}5$$

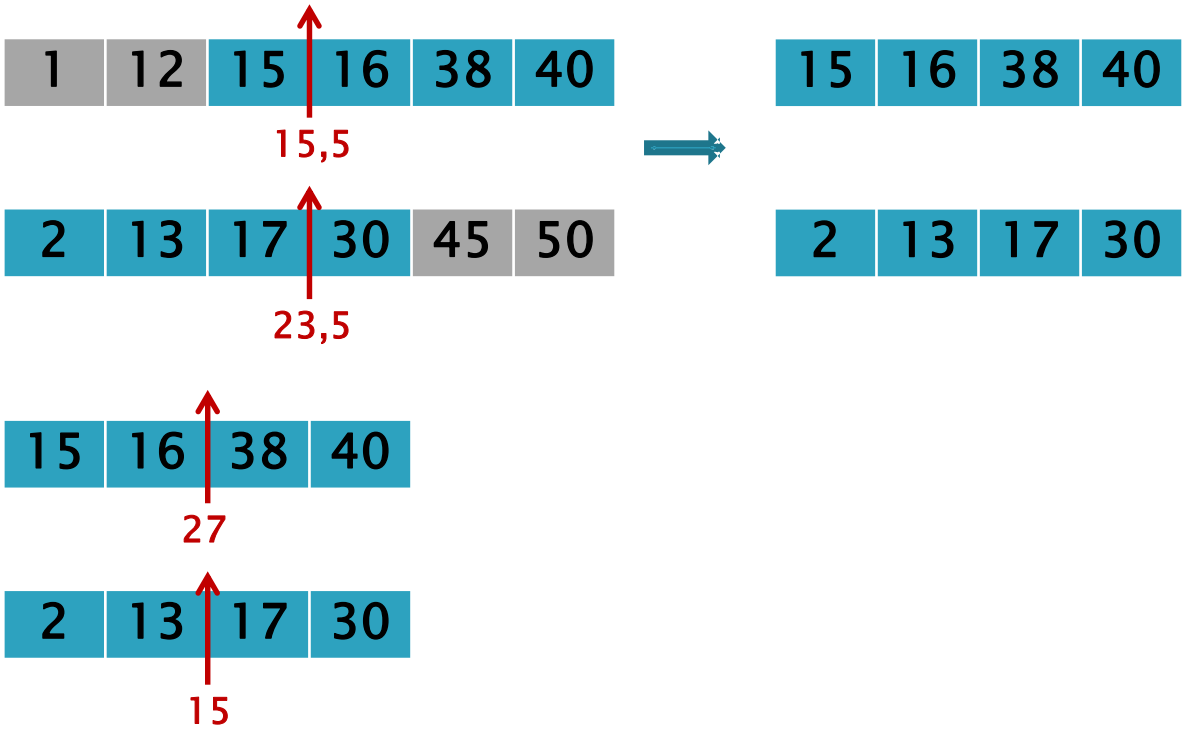
# Exemplul 2

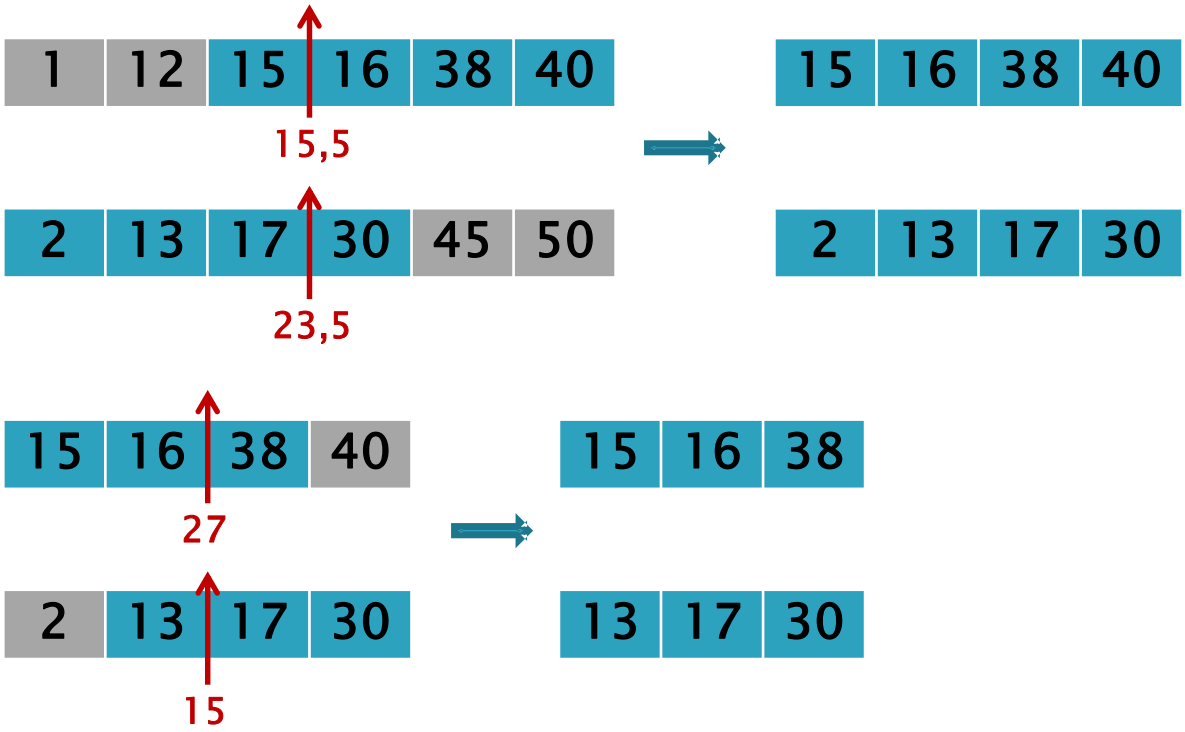
| 1 | 12 | 15 | 16 | 38 | 40 |
|---|---|---|---|---|---|

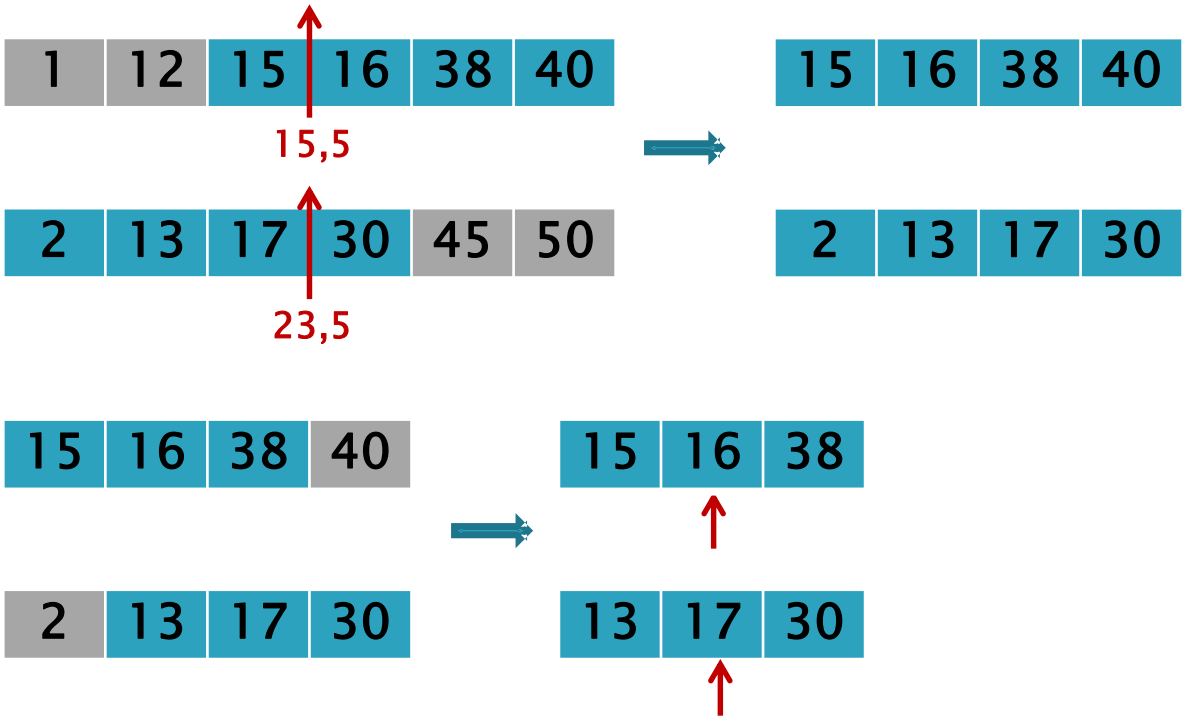
| 2 | 13 | 17 | 30 | 45 | 50 |
|---|---|---|---|---|---|

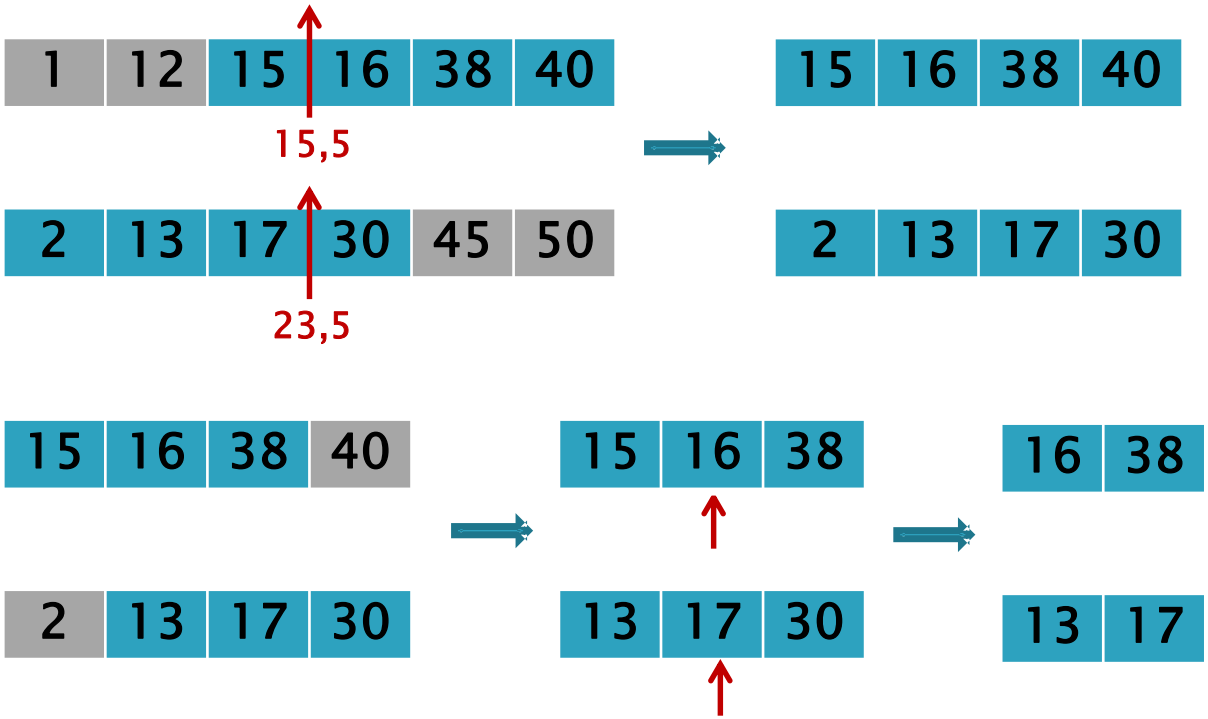

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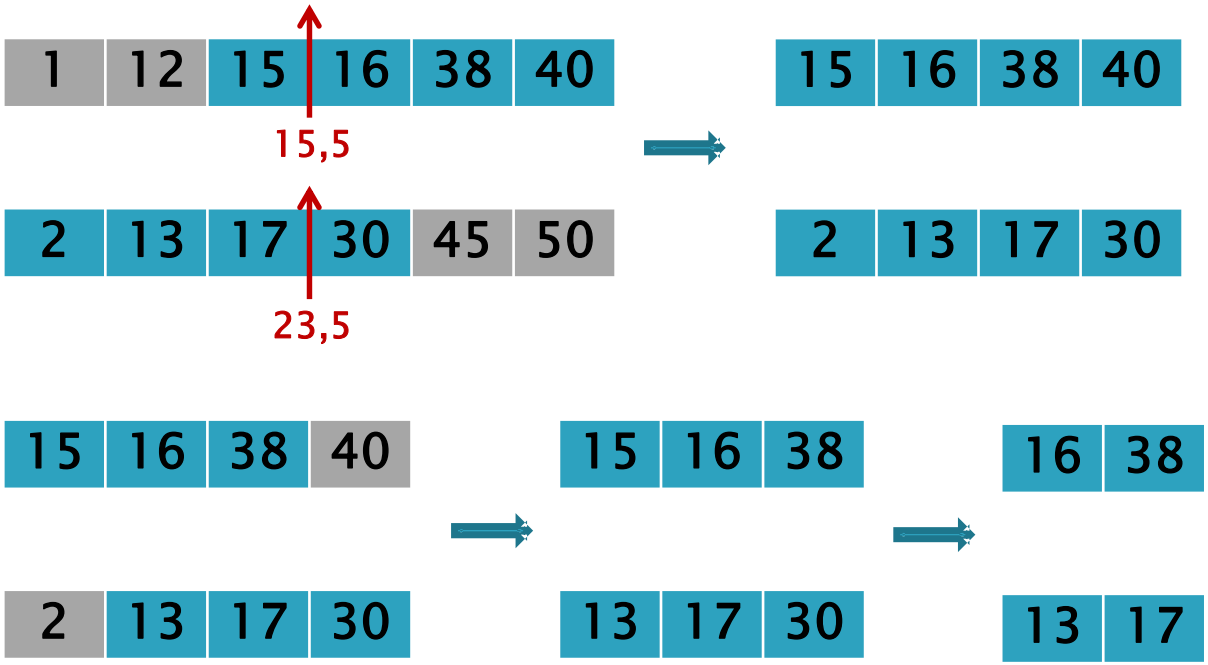

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1 12 15 16 38 40
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15 16 38
13 17 30
16 38
13 17




$$\text{Mediana} = \frac{\max\{13,16\}+\min\{17,38\}}{2} = \frac{16+17}{2}$$

$$= 16,5$$

# Algoritm

```
mediana(v, pv, uv){
      n = uv-pv+1
      m =(uv+pv)/2
      if (n%2==0)
         return(v[m]+v[m+1])/2
      else
         return v[m]
}
```

```
calculMediana(pa, ua, pb, ub){
    n = ua-pa+1
    if (n<=2) //rezolv direct
      return (max(a[pa],b[pb])+min(a[ua],b[ub]))/2
```



SAU

```
m = min(a[pa],a[ua],b[pb],b[ub])
M = max(a[pa], a[ua], b[pb], b[ub])
return (sum([a[pa],a[ua],b[pb],b[ub]]) - m - M)/2
```

```
calculMediana(pa, ua, pb, ub){
    n = ua-pa+1
    if (n<=2) //rezolv direct
      return (max(a[pa],b[pb])+min(a[ua],b[ub]))/2
    ma = mediana(a,pa,ua);//mediana lui a[pa..ua]
    mb = mediana(b,pb,ub);//mediana lui b[pb..ub]
    if(ma == mb) return ma
```

```
calculMediana(pa, ua, pb, ub){
    n = ua-pa+1
    if (n<=2) //rezolv direct
      return (max(a[pa],b[pb])+min(a[ua],b[ub]))/2
    ma = mediana(a,pa,ua);//mediana lui a[pa..ua]
    mb = mediana(b,pb,ub);//mediana lui b[pb..ub]
    if(ma == mb) return ma
    if (ma>mb)
      return calculMediana(pa, pa+n/2, pb+(n-1)/2,ub)
    else
      return calculMediana(pa+(n-1)/2, ua, pb,pb+n/2)
}

calculMediana(){
      return calculMediana(0,n-1,0,n-1)
}
```

# Mediana a doi vectori sortați

- **Complexitate:** O(log n)

# Mediana a doi vectori sortați



Mai este valabilă ideea pentru vectori de lungimi diferite (reducem problema la o problemă de acelaşi tip păstrând jumătate din fiecare vector)?

# Mediana a doi vectori sortați

Exemplu: n = 5

1 12 15 16 38

2 13 17 30 45



1 2 12 13 15 16 17 30 38 45

Mediana (15+16)/2 = 15,5